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Plot shapefile with matplotlib

Asked 9 years, 8 months ago Modified 3 years, 11 months ago Viewed 88k times



I am trying to read a shapefile and plot it using matplotlib. Here is the code:

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```
import matplotlib.pyplot as plt
import shapefile
```



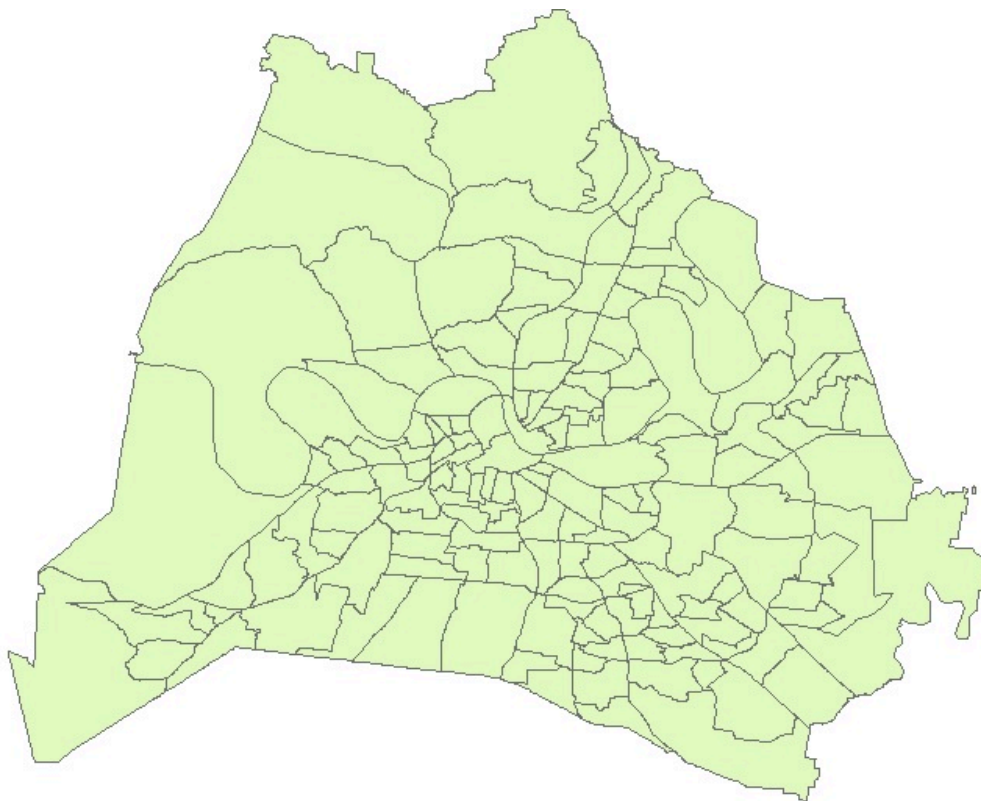
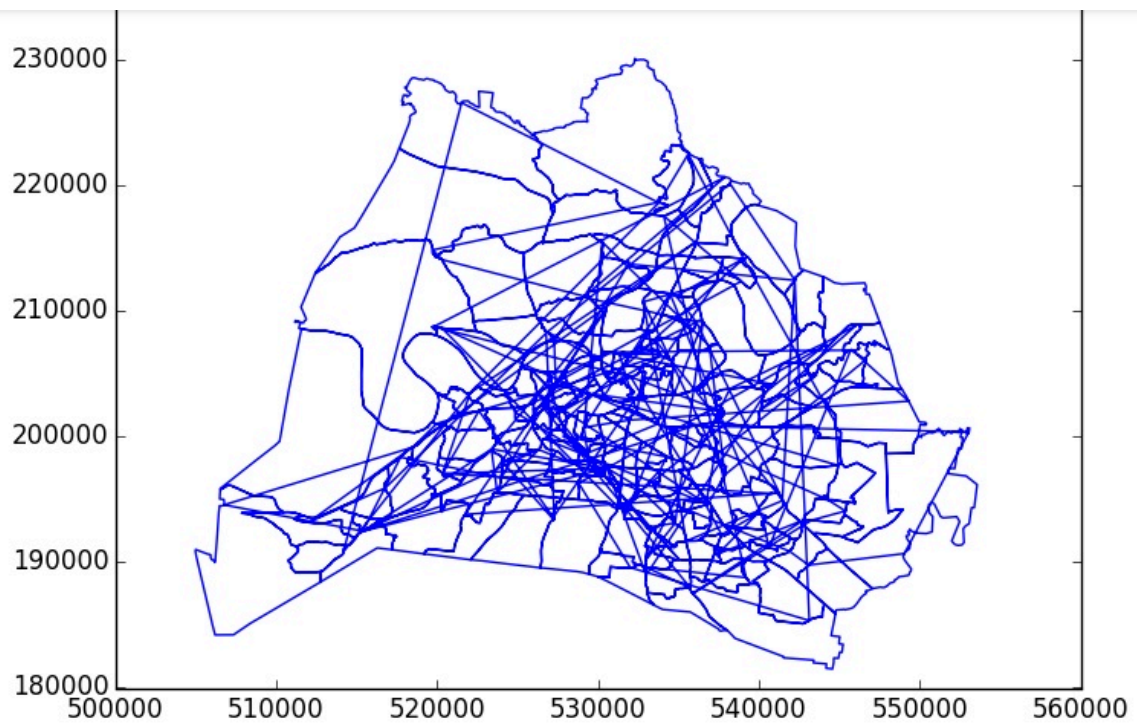
```
shpFilePath = "D:\test.shp"
listx=[]
listy=[]
test = shapefile.Reader(shpFilePath)
for sr in test.shapeRecords():
    for xNew,yNew in sr.shape.points:
        listx.append(xNew)
        listy.append(yNew)
plt.plot(listx,listy)
plt.show()
```



However, i get lines connecting my polygons. How can I draw the polygons such that they are the way in the shapefile. Here are screenshots of the plot and the shapefile when it is opened with ArcGIS.

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[shapefile](#)

[matplotlib](#)

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asked Jan 25, 2015 at 22:26



statBeginner

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of shapes to which you append each shapes points – user681 Jan 25, 2015 at 23:08

▲ Right. Have to find a way to separate the shapes. But that is what I am unable to do at the moment.
 ▼ – statBeginner Jan 25, 2015 at 23:24

▲ @DanPatterson Can you specify how to plot multiple shapes in the same figure after I manage to
 ▼ separate the shapes? If i use plt.plot(listx,listy) for every shape, it keeps generating a new figure
 every time, instead of using the same figure. – statBeginner Jan 26, 2015 at 0:12

7 Answers

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For future references, here is the solution I have came to after following the advices above.

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```
import shapefile as shp # Requires the pyshp package
import matplotlib.pyplot as plt
```



```
sf = shp.Reader("test.shp")
```



```
plt.figure()
for shape in sf.shapeRecords():
    x = [i[0] for i in shape.shape.points[:]]
    y = [i[1] for i in shape.shape.points[:]]
    plt.plot(x,y)
plt.show()
```

The resulting figure will be very colorful, but then, you just need to adjust the plot keywords.

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edited Oct 26, 2018 at 13:32

answered Jun 26, 2015 at 10:55



Martin Thoma

287 1 3 9



Idocao

191 1 3

9 ▲ I know this might be redundant information, but for those not yet familiar with the subject it would
 ▼ have been useful to say that `import shapefile` refers to the `pyshp` package:
pypi.python.org/pypi/pyshp – FaCoffee Feb 15, 2017 at 10:42

▲ This is not ok when you have a bunch of islands, as these points will be connected by lines to points
 ▼ on the mainland, creating something similar to what the OP posted. – FaCoffee Sep 28, 2017 at 11:21

2 ▲ @FaCoffee, you are right. My answer gis.stackexchange.com/a/309780/126618 should address this.
 ▼ – Gus Jan 24, 2019 at 17:07



I will leave it to you how to collect the shapes but this is the principle

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```
import numpy as np
from matplotlib.pyplot import plot #contains both numpy and pyplot
x1 = [-1,-1,10,10,-1]; y1 = [-1,10,10,-1,-1]
```

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p. show()



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answered Jan 26, 2015 at 0:20

user681



oh.. wonder how i missed that. i do get the shapes printed in different colors though. Will have to fix that :) – [statBeginner](#) Jan 26, 2015 at 5:46



how to get or isolate the different shapes? – [FaCoffee](#) Jun 21, 2016 at 10:29



You need to use matplotlib [paths](#) and [patches](#) and there is a Python module dedicated to plot polygons from shapefiles using these functions [Descartes](#).

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As Pyshp (shapefile) has the [geo interface](#) ([New geo interface for PyShp](#)) convention, you can use it.



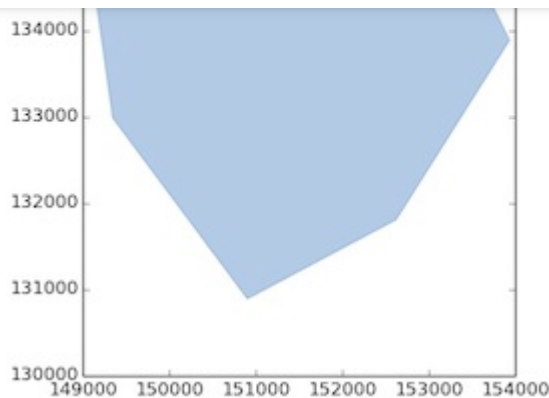
```
polys = shapefile.Reader("polygon")
# first polygon
poly = polys.iterShapes().next().__geo_interface__
print poly
{'type': 'Polygon', 'coordinates': (((151116.87238259654, 135890.8706318218),
(153492.19971554304, 134793.3055883224), (153934.50204650551,
133892.31935858406), (152623.97662143156, 131811.86024627919),
(150903.91200102202, 130894.49244872745), (149347.66305874675,
132991.33312884573), (149151.08424498566, 134383.76639298678),
(151116.87238259654, 135890.8706318218))),)}
```

The result is the GeoJSON representation of the geometry and you can use the solution of [How to plot geo-data using matplotlib/python](#)

```
import matplotlib.pyplot as plt
from descartes import PolygonPatch
BLUE = '#6699cc'
fig = plt.figure()
ax = fig.gca()
ax.add_patch(PolygonPatch(poly, fc=BLUE, ec=BLUE, alpha=0.5, zorder=2 ))
ax.axis('scaled')
plt.show()
```

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edited Apr 13, 2017 at 12:33

answered Jan 26, 2015 at 16:45



Community Bot
1



gene
55.3k 3 113 189

▲ That's really helpful, but can you do this in a for loop if you have multiple polygons to plot?
– FaCoffee May 15, 2017 at 13:10

▲ Yes without problem – gene May 15, 2017 at 15:39

▲ I noticed that the `descartes` solution does not work if you try and plot two different shapefiles on two adjacent subplots using `fig, ax = plt.subplots(1,2,figsize=(15, 8))` and then `ax[0].add_patch(PolygonPatch(poly_geo, fc='#d3d3d3', ec='#000000', alpha=0, zorder=5))` and `ax[1].add_patch(PolygonPatch(poly_geo, fc='#d3d3d3', ec='#000000', alpha=0, zorder=5))`. The result is an empty image. Any idea? – FaCoffee Jan 7, 2018 at 18:02



The most simple answer is using geopandas itself.

4

```
import geopandas as gpd
df=gpd.read_file("filename.shp")
df.plot()
```



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edited Oct 20, 2020 at 11:22

answered Oct 20, 2020 at 10:25



Kadir Şahbaz
77.8k 56 256 402



sameer_nubia
141 2



In addition to Idocao answer, and responding to FaCoffee question. When you have isolated islands and they are part of the same feature, you can try next:

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```
import shapefile as shp
import matplotlib.pyplot as plt

sf = shp.Reader("test.shp")
```



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```
i_end = len(shape.shape.points)
else:
    i_end = shape.shape.parts[i+1]
x = [i[0] for i in shape.shape.points[i_start:i_end]]
y = [i[1] for i in shape.shape.points[i_start:i_end]]
plt.plot(x,y)
plt.show()
```

This makes it work for me. The propertie "parts" of a shape returns the starting indexes of differents geometries inside a feature.

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answered Jun 19, 2018 at 22:04



felipesaam

61 2



Still, in one shapefile shape, there may be multiple parts. This will plot each part within one shape, separately.

3



```
import matplotlib.pyplot as plt
import shapefile
import numpy as np
```



```
this_shapefile = shapefile.Reader(map_file_base) # whichever file
shape = this_shapefile.shape(i) # whichever shape
points = np.array(shape.points)
```

```
intervals = list(shape.parts) + [len(shape.points)]
```

```
ax = plt.gca()
ax.set_aspect(1)
```

```
for (i, j) in zip(intervals[:-1], intervals[1:]):
    ax.plot(*zip(*points[i:j]))
```

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answered Jan 24, 2019 at 17:06



Gus

131 2



It can be done using either geopandas or pyshp as discussed in [this answer](#). Geopandas use matplotlib at its backend for plotting.

3



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answered Jan 2, 2018 at 6:51



MS_

171 3



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